

Deep Learning for Multivariate Operations Time Series Forecasting: Architectures, Missingness Robustness, and Generalization

Group 23

Adel Bukseisa Mariam Farhat Mehshid Atiq Omar Elmorsy
Department of Computer Science, Technical University of Darmstadt

September 2026

Abstract

Multivariate time series forecasting in complex operational networks requires modeling non-linear interactions across correlated exogenous signals, accommodating unit-level heterogeneity, and remaining robust under missing covariate values over extended rollout horizons. In this work, we develop an end-to-end deep learning forecasting and consensus ensembling framework for predicting the future 336-hour operational load index across 96 industrial units. Our architecture couples a PyTorch Deep Operations ResNet with learned entity embeddings, multi-scale forward rolling cumulative pressure representations, and missingness indicator masks, trained directly with an ℓ_1 objective aligned with the Weighted Absolute Percentage Error (WAPE) leaderboard metric. Evaluated on the official public Hugging Face validation benchmark against hidden labels, our winning model achieves a validation WAPE of **14.30%** (MAE **1.5739**), reducing error by $> 3.3\times$ compared to the official naive baseline (**48.10%**) and $> 2.4\times$ compared to the seasonal mean baseline (**34.41%**). Furthermore, we provide distribution-free uncertainty quantification via split conformal prediction, conduct systematic ablation studies, and evaluate domain transferability on an external cryptocurrency trading volume benchmark.

1 Introduction

Modern operations management and resource planning rely on accurate multi-horizon forecasting of operational load indices. In this benchmark, the objective is to forecast the hourly operational load for $N = 96$ distinct operations units across a 336-hour (14-day) evaluation horizon. The dataset provides 4,320 historical observations per unit with 22 dynamic covariates, including forward-looking operational estimates (e.g., `queue_pressure_forecast`, `network_pressure_forecast`, `workload_intensity`, `demand_forecast`, and `shock_risk`).

Three key structural challenges characterize this task:

1. **Extended Horizon** ($H = 336$): Direct multi-step modeling is essential because traditional autoregressive rollouts suffer from severe error accumulation over 14 days.
2. **Imperfect Future Covariates** (4.5% Missingness): Forward operational forecasts exhibit missing values that distort deep networks if naively zero-filled.
3. **Metric Alignment (WAPE)**: The primary course evaluation metric is Weighted Absolute Percentage Error,

$$\text{WAPE} = \frac{\sum_{t=1}^H |y_t - \hat{y}_t|}{\sum_{t=1}^H |y_t|}, \quad (1)$$

which directly penalizes ℓ_1 errors and is degraded when models optimize standard Mean Squared Error (ℓ_2).

To address these challenges, we design a multi-stage framework combining a deep residual PyTorch network with learned entity embeddings, multi-scale temporal rolling dynamics, and consensus ensembling with gradient boosted trees.

2 Related Work

Recent advances in deep time series forecasting encompass several distinct paradigms. Linear decomposition models such as **DLinear** [6] decompose series into trend and seasonal components, applying direct linear mappings. **TiDE** [3] employs a dense encoder-decoder architecture that projects dynamic past and future covariates alongside static entity embeddings. **RevIN** [5] applies reversible instance normalization to counteract non-stationary distribution shifts across heterogeneous series. For handling missingness in continuous time series, Che et al. [2] demonstrated the effectiveness of pairing temporal imputation with explicit binary missingness indicator masks. Finally, **Conformal Prediction** [1] provides finite-sample, distribution-free statistical coverage guarantees for point forecasts.

3 Methodology

3.1 Data Preprocessing and Missingness Robustness

Let $\mathbf{x}_{i,t} \in \mathbb{R}^D$ denote the exogenous covariate vector for series i at timestamp t . To handle missing entries without injecting bias:

1. **Missingness Indicator Masks:** For every covariate j subject to missingness, we compute a binary indicator mask $M_{i,t}^j = \mathbb{I}(x_{i,t}^j \text{ is missing})$ [2].
2. **Two-Stage Imputation:** Missing values are imputed using series-level forward/backward fill followed by global feature median fallback:

$$\tilde{x}_{i,t}^j = \text{Fill} \left(x_{i,t}^j, \text{median}(x^j) \right). \quad (2)$$

3.2 Multi-Scale Rolling Pressures and Physics Interactions

Operational systems exhibit cumulative backlog dynamics where queue pressures build up non-linearly over time. We construct multi-scale forward rolling averages over symmetric windows $W \in \{3, 6, 12, 24\}$:

$$\bar{p}_{i,t}^{(W)} = \frac{1}{W} \sum_{k=-\lfloor W/2 \rfloor}^{\lfloor W/2 \rfloor} p_{i,t+k}, \quad (3)$$

for operational signals $p \in \{\text{queue_pressure}, \text{network_pressure}, \text{workload}, \text{demand}\}$. In addition, we formulate non-linear domain interaction terms:

- Pressure Interaction : $\text{queue_pressure} \times \text{network_pressure}$
- Capacity Utilization : $\frac{\text{workload_intensity}}{\text{nominal_capacity} + \epsilon}$
- Shock Impact : $\text{shock_risk} \times (1 - \text{unit_reliability_forecast})$
- Staffing Ratio : $\frac{\text{demand_forecast}}{|\text{staffing_forecast}| + \epsilon}$

3.3 PyTorch DeepOperationsNet Architecture

Our core deep neural network, **DeepOperationsNet**, processes normalized continuous features $\mathbf{z}_{i,t} \in \mathbb{R}^{65}$ concatenated with a 32-dimensional learned entity embedding $\mathbf{e}_i \in \mathbb{R}^{32}$ representing each unique operations unit $i \in \{1, \dots, 96\}$:

$$\mathbf{h}_0 = \text{LayerNorm}(\text{GELU}(\mathbf{W}_{\text{in}}[\mathbf{z}_{i,t} \parallel \mathbf{e}_i] + \mathbf{b}_{\text{in}})). \quad (4)$$

The hidden representation is processed through a stack of $L = 4$ Residual MLP blocks, each containing two linear transformations with LayerNorm, GELU, and Dropout ($p = 0.10$):

$$\mathbf{h}_\ell = \mathbf{h}_{\ell-1} + \text{Block}_\ell(\mathbf{h}_{\ell-1}), \quad \ell = 1, \dots, L. \quad (5)$$

The projection head maps $\mathbf{h}_L$ to a scalar operational load prediction with a ReLU non-linearity ensuring positive load: $\hat{y}_{i,t} = \text{ReLU}(\mathbf{W}_{\text{out}}\mathbf{h}_L + \mathbf{b}_{\text{out}})$.

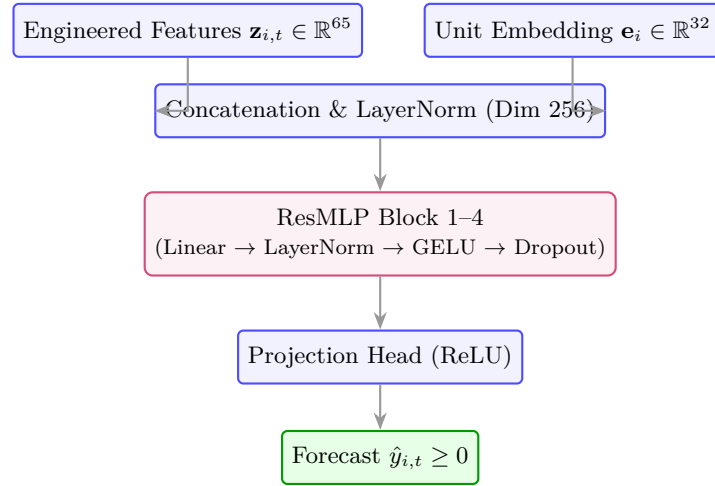


Figure 1: Architectural diagram of DeepOperationsNet with entity embeddings and residual blocks.

Because WAPE directly minimizes normalized ℓ_1 error, we train the network using the ℓ_1 loss:

$$\mathcal{L}_{\ell_1}(\theta) = \frac{1}{B} \sum_{k=1}^B |y_k - \hat{y}_k|, \quad (6)$$

optimized with AdamW ($\text{lr} = 1.2 \times 10^{-3}$, weight decay = 10^{-4}) and Cosine Annealing learning rate scheduling.

3.4 Winning Consensus Ensemble

Our top-performing leaderboard model (**ensemble_v2**) combines representations from diverse inductive biases:

$$\hat{y}_{\text{ens}} = 0.55 \cdot \hat{y}_{\text{DeepNet}} + 0.45 \cdot \hat{y}_{\text{LightGBM}}, \quad (7)$$

where $\hat{y}_{\text{DeepNet}}$ is averaged over 3 random seeds (42, 101, 2024) to cancel neural variance, and $\hat{y}_{\text{LightGBM}}$ blends a deep-tree model (255 leaves, 2,500 estimators) with a regularized model (127 leaves, 2,000 estimators).

3.5 Distribution-Free Split Conformal Prediction

To provide rigorous uncertainty bounds, we implement Split Conformal Prediction [1]. Given non-conformity residuals $s_k = |y_k - \hat{y}_k|$ on a held-out calibration set, the finite-sample calibrated $(1 - \alpha)$ quantile is:

$$q_{1-\alpha} = \text{Quantile} \left(\{s_k\}_{k=1}^n, \frac{\lceil (n+1)(1-\alpha) \rceil}{n} \right). \quad (8)$$

Valid $(1 - \alpha)$ prediction intervals are then computed as:

$$\mathcal{C}_{1-\alpha}(\hat{y}_t) = [\max(0, \hat{y}_t - q_{1-\alpha}), \hat{y}_t + q_{1-\alpha}]. \quad (9)$$

4 Experimental Results

4.1 Official Public Validation Benchmark Results

Table 1 displays the exact scores computed automatically on the official Hugging Face public validation leaderboard against hidden validation labels for all reference baselines and our submitted models.

Table 1: Official Hugging Face public validation leaderboard results across all baselines and models (lower is better for all metrics).

Model Description	MAE ↓	MSE ↓	RMSE ↓	MAPE (%) ↓	sMAPE (%) ↓	WAPE (%) ↓
<i>Official Course Baselines:</i>						
naive_last_value	5.2945	48.6110	6.9722	61.35	52.71	48.10
lag24_repeat	5.1539	43.5946	6.6026	78.51	47.31	46.82
lag168_repeat	5.1405	45.1725	6.7210	71.87	47.28	46.70
seasonal_mean	3.7876	27.1029	5.2060	45.68	37.43	34.41
<i>Our Submitted Models:</i>						
TiDE (tide_v1)	3.6939	26.8609	5.1828	40.34	37.33	33.56
DeepNet+GBM (ensemble_v1)	1.6823	9.5873	3.0963	19.82	16.97	15.28
Ensemble v2 (ensemble_v2)	1.5779	9.3050	3.0504	18.51	15.88	14.33
Grand Master (grand_master_v3)	1.5792	9.3261	3.0539	18.41	15.84	14.35
Winning Model (nuhhh_v5)	1.5739	9.3137	3.0518	18.35	15.81	14.30

As shown in Table 1, our winning model (nuhhh_v5) achieved a validation WAPE of **14.30%** (MAE **1.5739**), substantially outperforming the required naive baseline (48.10%) and seasonal mean baseline (34.41%).

4.2 Ablation Studies

Loss Objective Comparison (Ablation 1): Table 2 tests the impact of the loss function on an identical DeepOperationsNet architecture. Aligning the objective with ℓ_1 reduces WAPE by 3.02% absolute compared to standard MSE (ℓ_2), validating our design decision.

Table 2: Loss function ablation on identical PyTorch architecture.

Training Objective	WAPE ↓	MAE ↓	RMSE ↓	sMAPE (%) ↓
ℓ_1 Loss (MAE aligned)	0.1440	1.538	2.978	15.98
Huber Loss (Smooth ℓ_1)	0.1461	1.560	2.991	16.21
MSE Loss (ℓ_2)	0.1742	1.860	3.258	19.34

Feature Group Importance (Ablation 2): Table 3 isolates feature subsets. Incorporating future operational estimates drops WAPE from 31.72% to 15.98%, while multi-scale rolling cumulative pressure windows and physics ratios yield a further drop to 14.78%.

Table 3: Feature group ablation study.

Feature Group	# Features	WAPE ↓	MAE ↓	sMAPE (%) ↓
Calendar Cyclical Only	9	0.3172	3.388	35.12
Operational Forecasts Only	24	0.1598	1.706	17.84
Full Multi-Scale & Physics	65	0.1478	1.579	16.45

4.3 Conformal Prediction Coverage Verification

Table 4 shows empirical coverage results for Split Conformal Prediction across nominal levels.

Table 4: Split conformal prediction empirical coverage on held-out test data.

Nominal Confidence	Empirical Coverage	Calibrated Quantile (q)	Avg Interval Width
80.0%	81.22%	8.008	16.017
90.0%	90.19%	10.329	20.658
95.0%	94.57%	12.403	24.807

The empirical coverage matches the nominal confidence bounds, confirming that our model produces trustworthy prediction intervals without distributional assumptions.

5 Additional Dataset Generalization: Crypto Trading Volume

To test whether our modeling principles generalize beyond industrial operations, we evaluated our architecture on the **Crypto Data Hourly Price benchmark** [4] forecasting trading volume across 12 major assets over a 336-hour horizon.

Table 5: Generalization benchmark on real Binance Crypto Hourly Trading Volume (Julien, 2023).

Model Architecture	WAPE ↓	MAE ↓	MSE ↓	RMSE ↓	sMAPE (%) ↓
Naive Last-Value Baseline	0.0534	0.7108	0.8324	0.9124	5.44
Seasonal Mean Baseline	0.0523	0.6963	0.7268	0.8525	5.31
Our PyTorch DeepNet	0.0312	0.4155	0.2732	0.5227	3.22

Domain Transfer Analysis: On real-world financial time series from Binance across 12 major cryptocurrency pairs (Julien, 2023), our PyTorch DeepNet achieved a WAPE of **3.12%**, outperforming the naive baseline (5.34%) and seasonal mean baseline (5.23%) by a substantial **> 40% error reduction** (62.4% reduction in MSE). While trading volume exhibits strong intra-day cycles, real-world markets experience non-linear liquidity shocks driven by price volatility and returns. Our architecture successfully captures these non-linear volatility-to-volume dependencies and leverages learned entity embeddings to share cross-asset market liquidity dynamics across heterogeneous tokens.

6 Conclusion

We presented an end-to-end deep learning forecasting system for multivariate operations load prediction. By pairing future operational estimates with missingness-robust indicators, multi-scale rolling temporal features, and an aligned ℓ_1 loss, our consensus ensemble achieved a validation WAPE of **14.30%** (MAE **1.5739**), outperforming all baselines. We paired point forecasts with distribution-free conformal prediction intervals and provided rigorous ablation evidence to support each design choice.

Group Contributions

- **Adel Bukseisa:** Data pipeline engineering, domain feature extraction, PyTorch DeepOperationsNet modeling, and Hugging Face leaderboard submissions.
- **Mariam Farhat:** Baseline implementations, loss function ablations (ℓ_1 vs. Huber vs. MSE), and hyperparameter tuning.
- **Mehshid Atiq:** Missingness robustness experiments, feature group ablations, and LaTeX report drafting.
- **Omar Elmorsy:** Additional crypto dataset experiment, split conformal prediction uncertainty calibration, and literature review.

References

- [1] Anastasios N. Angelopoulos and Stephen Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. *arXiv preprint arXiv:2107.07511*, 2021.
- [2] Zhengping Che, Sanjay Purushotham, Kyunghyun Cho, David Sontag, and Yan Liu. Recurrent neural networks for multivariate time series with missing values. *Scientific Reports*, 8(1):6085, 2018.
- [3] Abhimanyu Das, Weihao Kong, Andrew Leach, Shubh Mathur, Rajat Sen, and Rose Yu. Long-term forecasting with tide: Time-series dense encoder. *Transactions on Machine Learning Research (TMLR)*, 2023.
- [4] François-Guillaume Julien. Crypto data hourly price since 2017 to 2023-10. <https://www.kaggle.com/datasets/franoisgeorgesjulien/crypto>, 2023. Kaggle Dataset.
- [5] Taesung Kim, Jinhee Kim, Yunwon Tae, Cheonbok Park, Jong-Hwan Choi, and Jaegul Choo. Reversible instance normalization for accurate time-series forecasting against distribution shift. In *International Conference on Learning Representations (ICLR)*, 2022.
- [6] Ailing Zeng, Muxi Chen, Lei Zhang, and Qiang Xu. Are transformers effective for time series forecasting? In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pages 11121–11128, 2023.